

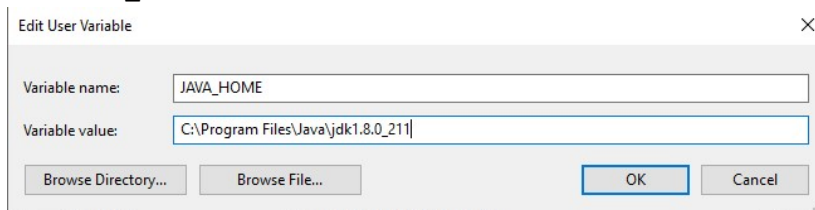
Setup Python Virtual Environment to run PySpark in VSCode

Install Prerequisites

1. **Make sure you are running Java 8 (JDK 1.8.xxx) or up. Install if required.**

- `java -version`
- URL: <https://www.oracle.com/in/java/technologies/downloads/#jdk24-windows>

2. **Add JAVA_HOME as an environment variable**



3. **Make sure a suitable version of python is installed on your machine**

- a. Check version compatibility using the following link
- b. <https://community.cloudera.com/t5/Community-Articles/Spark-Python-Supportability-Matrix/ta-p/379144>

Setup Python Virtual Environment (venv)

4. **Open command prompt (or PowerShell) on your windows machine and run the following commands to create a python virtual environment in your project folder.**

- a. Create a project folder in which you want to install the venv (virtual environment)
 - `mkdir pyspark-project`
- b. Create a python virtual environment (folder: **pyspark-project**, venv name: **venv**)
 - `cd pyspark-project`
 - `python -m venv venv`
- c. Activate the virtual environment.
 - `venv\Scripts\Activate.ps1`

5. **Install pyspark in the virtual environment and launch VSCode**

- a. Install PySpark using pip in your virtual environment
 - `pip install pyspark`

b. Launch VSCode

- `code .`

```
E:\PySpark\vscode>python -m venv venv  
  
E:\PySpark\vscode>venv\Scripts\activate.bat  
  
(venv) E:\PySpark\vscode>pip install pyspark  
  
(venv) E:\PySpark\vscode>code .
```

6. **Open your PySpark scripts in VSCode and execute the program.**

Note: If you are running the scripts provided by me, make sure you delete the first two lines in the code related to importing and initializing `findspark` library.

- a. You can run the python script in the VSCode code terminal using the following command:
 - i. `python <script-path>`
- b. If you want to run specific lines of python code, you can select the code and use Shift+Enter keyboard shortcut.

Common Issues & Fixes

Issue	Fix
<code>JAVA_HOME</code> not set error	Recheck Step 2, restart VS Code fully
<code>winutils.exe</code> / Hadoop errors on Windows	Download <code>winutils.exe</code> matching your Hadoop version, place in <code>C:\hadoop\bin</code> , set <code>HADOOP_HOME=C:\hadoop</code>
Pylance can't resolve <code>pyspark</code> import	Confirm correct interpreter selected (Step 7); reload window (Ctrl+Shift+P → Reload Window)
PowerShell activation blocked	Run the <code>Set-ExecutionPolicy</code> command from Step 3